
IPEDS DATA QUALITY AUDIT

2024 SURVEY CYCLE – FIVE DATASET REVIEW

FRAMEWORK: DAMA DMBOK SIX QUALITY DIMENSIONS

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AUDIT SCOPE & APPROACH

- Comprehensive data quality audit of five Integrated Postsecondary Education Data System (IPEDS) datasets published by the US Dept of Education
- IPEDS is the primary source of data on 6,000+ US higher education institutions
- Framework: DAMA DMBOK six data quality dimensions
- Tool: Python in Google Colab

Dataset	Description	Rows	Columns	Unique Institutions
df1	12-Month Enrollment	115,026	72	5,861
df2	Financial Data	1,911	288	1,911
df3	Graduation Rates	51,011	66	3,684
df4	Institution Directory	6,072	72	6,072
df5	Institution Characteristics	5,963	111	5,963

AUDIT FRAMEWORK: DAMA DMBOK

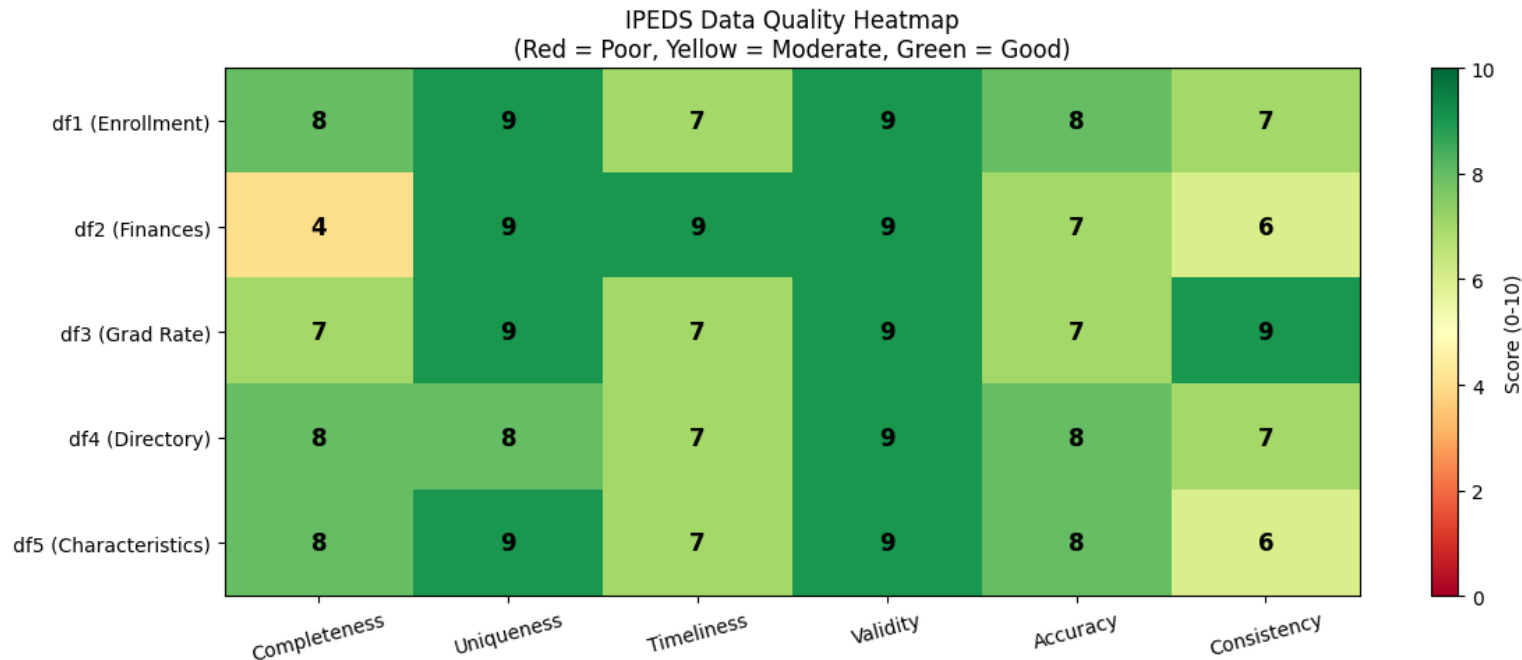
Six Data Quality Dimensions:

1. Completeness – Are all required values present?
2. Uniqueness – Are records free of unintended duplicates?
3. Timeliness – Is data current for its intended use?
4. Validity – Do values conform to defined rules and formats?
5. Accuracy – Does data correctly represent reality?
6. Consistency – Is data coherent across datasets?

Each dataset scored 0 – 10 per dimension

Based on DAMA International, Data Management Body of Knowledge

OVERALL DATA QUALITY RESULTS



Dataset	Score	Rating
df1 Enrollment	8.0	Good
df2 Finances	7.3	Acceptable
df3 Grad Rate	8.0	Good
df4 Directory	7.8	Acceptable
df5 Characteristics	7.8	Acceptable

Key Insights

No dataset scored below Acceptable
IPEDS data is well-structured and
suitable for research and policy analysis
with appropriate domain knowledge

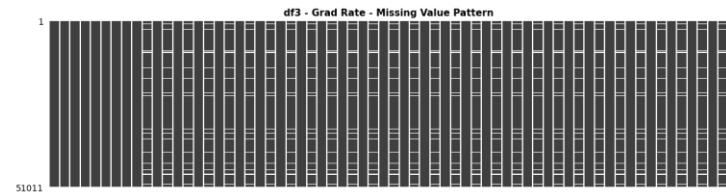
DIMENSION I: COMPLETENESS



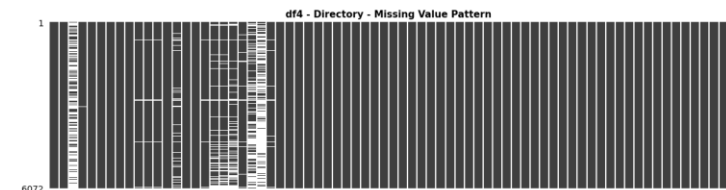
df1 Enrollment: Rare reporting situation columns 88-98% null, expected by design



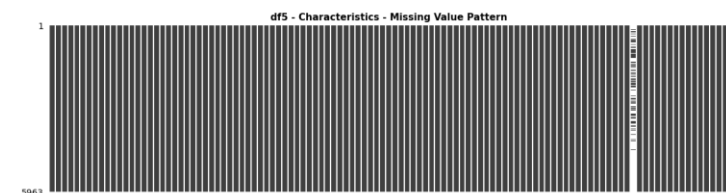
df2 Finances: 1,911 of 6,072 institutions report financial data, FIA covers public GASB institutions only. 97 columns contain up to 58% missing values



df3 Grad Rate: 3,684 of 6,072 institutions report graduation data, non-degree seeking institutions do not report. All 30 demographic columns have missing count of 3,623 rows



df4 Directory: Optional URL fields 14-72% null



df5 Characteristics: DISABPCT 71.2% null, only reported when rare threshold met

DIMENSION 2: UNIQUENESS

Strong results across all datasets

- Zero fully duplicated rows in any dataset
- df2, df4, df5: Zero duplicated UNITIDs
- df1, df3: Expected duplicate UNITIDs by design

Concerns

- 39 duplicate OPEIDs found
 - OPEIDs should be campus-specific and not shared across organization
 - Poses financial aid compliance risk

Key Insights

- 100% referential integrity
- Zero orphaned records across all datasets
- UNITID functions as reliable primary key across IPEDS system

DIMENSION 3: TIMELINESS

Institutions Closed in 2024 reporting period

71 Institutions reported Closed in df4 Directory

Dataset	# institutions still reporting
df1 Enrollment	14
df2 Finances	0
df3 Grad Rate	3
df5 Characteristics	17

Critical Data Gap

- No internal year column in df1 Enrollment or df2 Finances datasets
- Reporting year exists only in filenames
- Governance risk for longitudinal analysis
- Recommend IPEDS system-wide addition of internal year columns

DIMENSION 4: VALIDITY

- **Strongest dimension across all datasets**
- **Validity scores capped at 9 as checks were thorough, but not exhaustive**

Checks Performed	Results
Zip code formats	100% valid
State abbreviations	100% valid
ICLEVEL codes	All values -3,1,2,3
CONTROL codes	All values 1,2,3
Enrollment values	No decimals found
Graduation rate values	No negatives found
Character formats	Df4 UNITID had BOM, rendering as ï»¿UNITID

DIMENSION 5: ACCURACY

Enrollment Outliers

Top 10 institutions dominated by online universities

Top 10 Unique Institutions by Total Enrollment:				
	UNITID	EFYTOTLT	INSTNM	STABBR
0	433387	282150	Western Governors University	UT
1	183026	255134	Southern New Hampshire University	NH
2	150987	178903	Ivy Tech Community College	IN
3	104717	170467	Grand Canyon University	AZ
4	484613	156956	University of Phoenix-Arizona	AZ
5	232557	143337	Liberty University	VA
6	227182	107793	Lone Star College System	TX
7	224615	104064	Dallas College	TX
8	483124	98316	Arizona State University Digital Immersion	AZ
9	449339	95158	American Public University System	WV

Financial Outliers

System universities consolidate assets, Haskell reports under federal standards

Top 5 Institutions by Total Assets:				
	UNITID	F1A06	INSTNM	
0	124557	104112000000.00	University of California-System Administration...	
1	229090	92193294888.00	The University of Texas System Office	
2	170976	33755911000.00	University of Michigan-Ann Arbor	
3	204796	22436262359.00	Ohio State University-Main Campus	
4	236948	18488640118.00	University of Washington-Seattle Campus	
Institutions with Zero Total Assets: 1				
	UNITID	F1A06	INSTNM	
0	155140	0.00	Haskell Indian Nations University	

Geographic Outliers

7 flagged institutions are all US-affiliated Pacific Island institutions with accurate coordinates

Institutions with unexpected coordinates:				
	INSTNM	LONGITUD	LATITUDE	STABBR
3562	Guam Community College	144.81	13.44	GU
3563	University of Guam	144.80	13.43	GU
3564	Northern Marianas College	145.72	15.15	MP
3630	College of Micronesia-FSM	158.16	6.91	FM
3631	Palau Community College	134.47	7.34	PW
3848	College of the Marshall Islands	171.38	7.10	MH
4291	Pacific Islands University	144.84	13.47	GU

DIMENSION 6: CONSISTENCY

Key Insights

- Most complex dimension
- Domain knowledge required to analyze datasets
- Without IPEDS data dictionary, negative values appear to be errors
- Negative coding conventions are inconsistently defined in IPEDS dictionaries by dataset
- Results in reduced interoperability for analysts unfamiliar with conventions

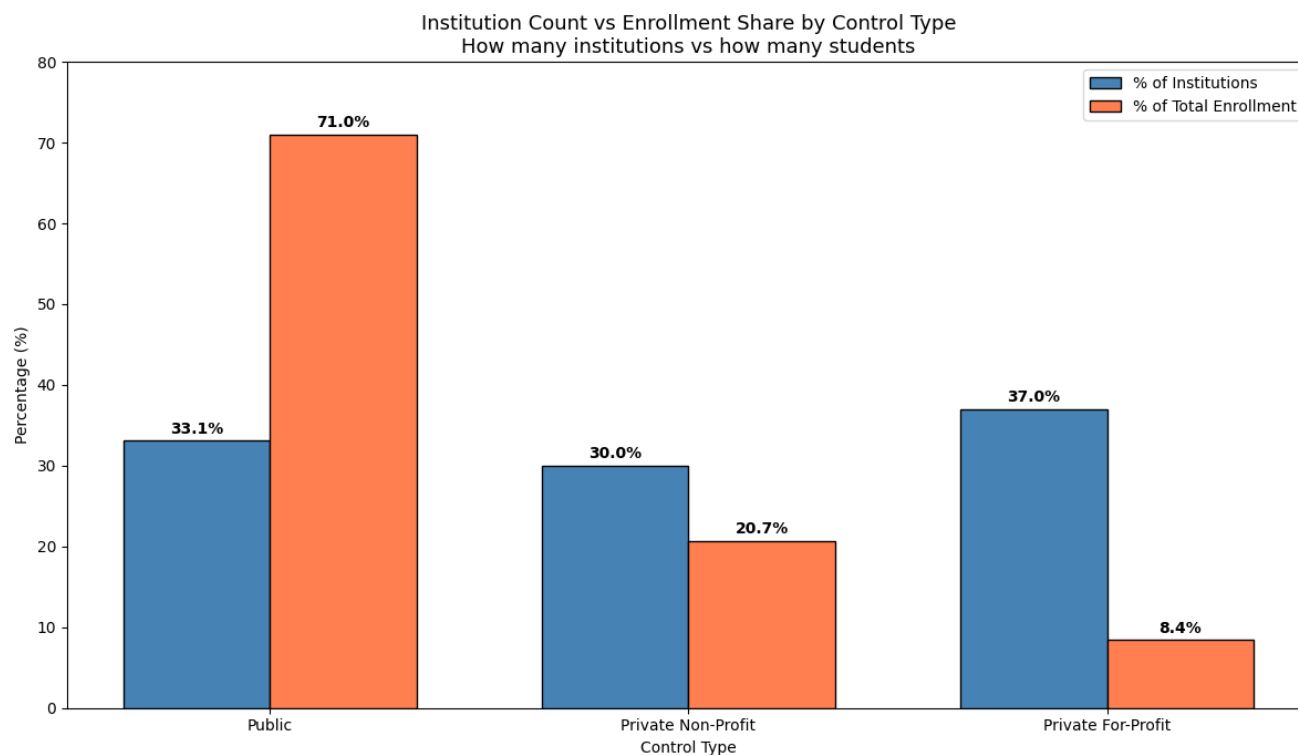
Negative Coding Conventions

-1	Not reported
-2	Not applicable
-3	Not available
-9	Not active

Dataset	# of columns affected
df1 Enrollment	1*
df2 Finances	41
df3 Grad Rate	0
df4 Directory	37
df5 Characteristics	95

*This one enrollment column contains 88% negative values

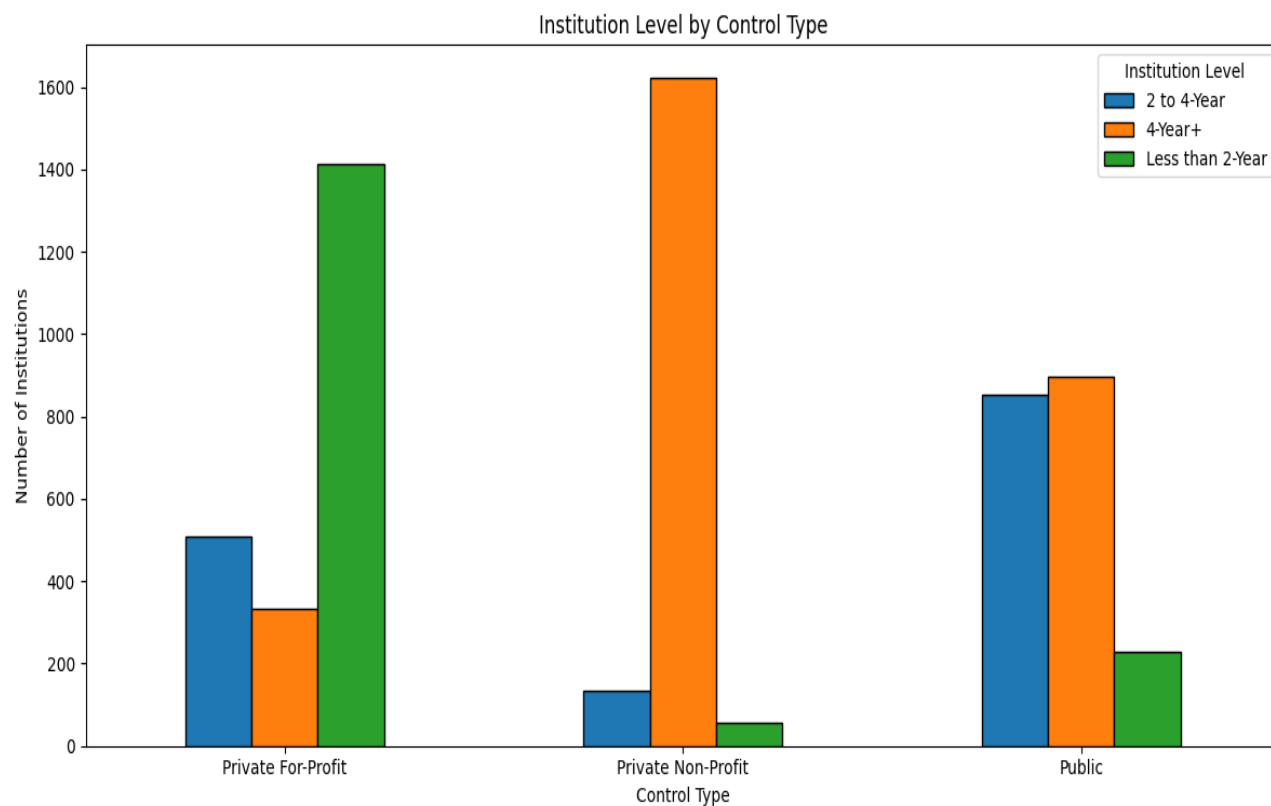
WHAT THE DATA REVEALS: INSTITUTIONAL LANDSCAPE



Key Insights

- For-profit institutions are the most numerous sector, but serve only 8.4% of students
- Public institutions represent one-third of institutions, but serve almost three-fourths of students
- Using institution counts as a proxy for educational impact will significantly overestimate the role of for-profit institutions and underestimate the dominance of public higher education

WHAT THE DATA REVEALS: SECTOR STRUCTURE

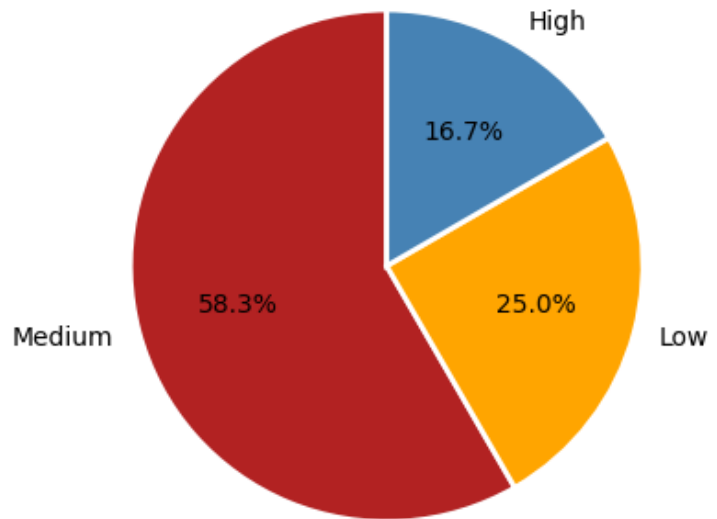


Key Insights

- **For-profit:** Largely consists of Less than 2-Year institutions providing vocational and certificate programs
- **Private Non-profit:** Almost entirely 4-Year+ institutions consisting of traditional degree-granting colleges and universities
- **Public:** More balanced distribution due to 4-Year university and community college institutions
- Cross-sector comparisons must account for these structural differences to arrive at meaningful results

FINDINGS MAPPED TO GOVERNANCE FRAMEWORK

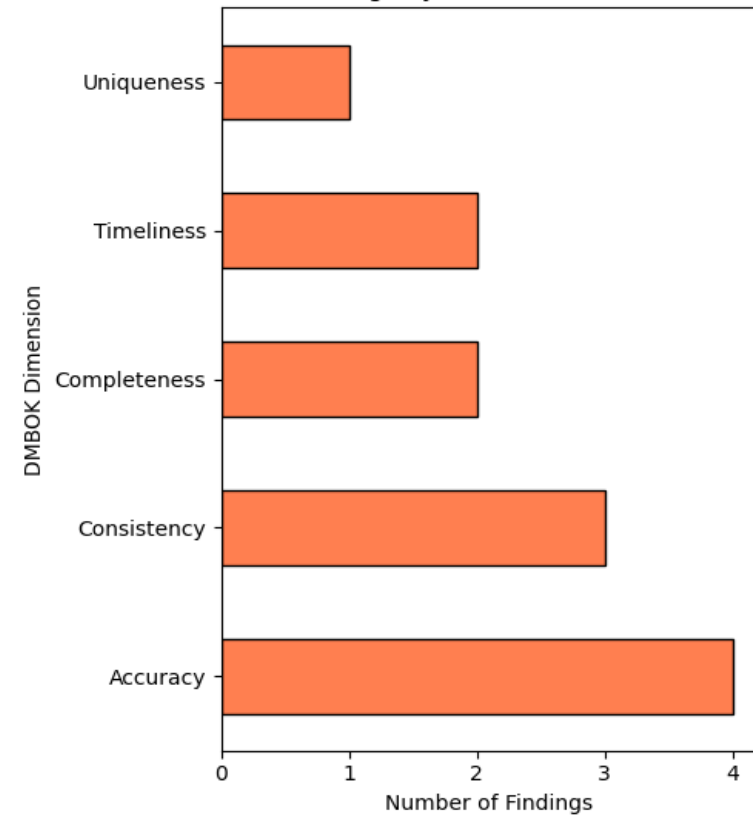
Findings by Severity



12 total findings documented across 5 of the 6 DAMA DMBOK data quality dimensions

- High: 2
- Medium: 7
- Low: 3

Findings by DMBOK Dimension



Relatively few data quality issues were found, with zero findings in the Validity dimension

PRIORITY RECOMMENDATIONS

Critical

- Only use dataset F1A alone when analyzing financial data for public universities. Combine with F2 and F3 (submitted by private institutions) for complete financial coverage

Medium

- Create centralized sentinel value registry for all negative coding conventions
- Apply DEATHYR filter before all cross-dataset analysis
- Standardize all IPEDS file encoding to UTF-8-SIG

High

- Review and correct shared OPEIDs
- Add REPORT_YEAR column to all datasets for longitudinal analysis safety

Low

- Add territory flag for Pacific Island institutions
- Always pair institution counts with enrollment figures to avoid misleading conclusions

KEY TAKEAWAYS

1. IPEDS data is well-structured and suitable for research, no dataset scored below Acceptable
2. Domain knowledge is required, most apparent anomalies are valid conventions requiring access to IPEDS data dictionary to interpret correctly
3. The finance file (F1A) covers 31.5% of institutions, files F2 and F3 are needed to bridge this governance gap
4. UNITID referential integrity is exceptional with a 100% match across all five datasets
5. Counting institution types and counting students tell opposite stories
6. Consider AI-assisted tools to overcome required human domain expertise and to rapidly surface the structural findings needed for deeper governance insights